Amiri Hayes

Machine Learning & Artificial Intelligence Research
akh5@njit.edu | www.amirihayes.com | 201-745-0886 | Cambridge, MA

EDUCATION:

NJIT | New Jersey Institute of Technology Sep 2023 - Dec 2026
MS Artificial Intelligence & BS Applied Mathematics (with Honors) [Full Ride Merit Scholarship]

Relevant Courses: Deep Learning, Natural Language Processing, Stochastic Simulation, Reinforcement Learning, Computer Vision, Numerical & Statistical Methods, Linear Algebra, Real Analysis I & II, Probability Distributions

RCSJ | Rowan College of South Jersey Triple Major: AS Physics, AS Mathematics, AS Computer Science

RECENT WORK:

- *Amiri Hayes*, Belinda Li, Jacob Andreas. **Explaining Attention with Program Synthesis.** NeurIPS Workshop 2026. Project: Established precedent for replacing LLM attention heads with a dataset of generalizable executable programs, which causally provides insight into logic at the attention stage while maintaining model performance.
- *Amiri Hayes*, Kristina Wicke. **Systematic Construction of Text-Attributed Graphs.** (ICLR 2027, under review) Project: Developed system to train decision trees based on text-attributed graph construction choices, such that construction choices can be systematically ranked for expected performance ahead of GNN and GraphRAG tasks.

RESEARCH EXPERIENCE:

linkedin.com/in/amirihayes – github.com/AmiriHayes

MIT CSAIL | LINGO, Language & Intelligence Group Cambridge, MA | MIT
Fall 2026 - Current · Pre-Doctoral Researcher · Mentors: Jacob Andreas & Belinda Li

- Year-long lab member focused on converting LLM activations to Python for causal mechanistic interpretability

Summer 2025 · Machine Learning Researcher, MSRP Intern

- Engineered Python interpretability library ViewLLM using program synthesis to classify LLM attention heads
- Built distributed Python & SLURM pipelines on CSAIL H100 clusters for scalable experiments and data flow
- Classified LLM activations & validated thousands of hypothesis functions, advancing attention head understanding

IBM Research | Thomas J. Watson Research Center Yorktown Heights, NY | IBM
Summer 2026 · Machine Learning Researcher, GEM Fellow Awardee · Mentors: John Smith & Anna Paola Carrieri

- Developed diffusion-based models to solve combinatorial optimization problems over graphs and hypergraphs
- Trained supervised, unsupervised, and quantum solvers applied to Max3SAT & MaxCut hypergraph instances
- Studied applicability of our theoretical speedups on finding solutions to binarized weights in neural networks

Brookhaven National Laboratory | Environmental Sciences & Technologies Upton, NY | BNL
Spring 2026 · Machine Learning Researcher, SULI Intern · Mentor: Min Deng

- Engineered statistical framework to identify cold fronts from ~100GB of climate data with 90%+ inference accuracy
- Developed spatiotemporal inference models to extrapolate event lifecycles & identify atmospheric transition timing
- Conducted correlation analysis of variables; saved results and loaded data via GCP to optimize loading performance

UCLA | Institute for Pure & Applied Mathematics Los Angeles, CA | UCLA
Summer 2024 · Machine Learning Researcher, RIPS Intern · Mentors: Susana Serna & Raffaele Vardavas

- Led team of 3 undergrads as project manager developing generative models for epidemiological network simulation
- Implemented VAE, GAN, and score-based algorithms w/ parallel computing to generate large social graph datasets
- Generated synthetic 100-person social networks with vaccination statistics, developed a network fusion technique

RELEVANT SKILLS:

Specializations: Deep Learning, Data Engineering, AI Safety, Computer Vision, Language Models
Certifications: Google Data Analytics, Cornell Break Through Tech AI, NJ Governor's STEM Scholar
Technical: Python, PyTorch, JavaScript, SQL, C++, AWS, Google Cloud, MongoDB, HuggingFace